Dept. of Advanced Computing Sciences, Maastricht University

# MATSE: Operation Research

## Lecture 9. Game theory - Applications

# Why Game Theory

- Provides the mathematical tools to ensure successful cooperation among users / players
- Is as a tool to design an algorithm
- Set rules of the game that will bring the interaction to the maximum global welfare.
- Model situations in which decision-makers interact with each other
- The happiness of each participant depends also on the decisions made by everyone

# Nash Equilibrium Problem

## Energy Market

- Set of agents: $\mathcal{I} = \{1, \dots, N\}$

  $\Rightarrow$ Companies

- Decision variable: $x_i \in \Omega_i$

  $\Rightarrow$ Quantity of Energy

- Objective Function: $J_i(x_i, \boldsymbol{x}_{-i})$

  $\Rightarrow$ Cost - Profit

# Nash Equilibrium Problem

## Energy Market

- Set of agents: $\mathcal{I} = \{1, \ldots, N\}$

  $\Rightarrow$ Companies

- Decision variable: $x_i \in \Omega_i$

  $\Rightarrow$ Quantity of Energy

- Objective Function: $J_i(x_i, \boldsymbol{x}_{-i})$

  $\Rightarrow$ Cost - Profit

$$\forall i \in \mathcal{I} \quad \begin{cases} \min & J_i(x_i, \boldsymbol{x}_{-i}) \\ \text{s.t.} & x_i \in \Omega_i \end{cases}$$

# Nash Equilibrium

A Nash equilibrium is a collective strategy $\boldsymbol{x}^* \in \Omega$ such that for all $i \in \mathcal{I}$

$$J_i(x_i^*, \boldsymbol{x}_{-i}^*) \leq J_i(y, \boldsymbol{x}_{-i}^*) \text{ for all } y \in \Omega_i.$$

J. F. Nash, Equilibrium points in n-person games, Proceedings of the national academy of sciences, (1950), 36(1), 48-49.

# Nash Equilibrium

## An example

*Cost Matrix*

Company 1

| Company 2 | | $M_1$ | $M_2$ |
|---|---|---|---|
| | $M_1$ | 4 , 4 | 3 , 2 |
| | $M_2$ | 2 , 3 | 1 , 1 |

# Nash Equilibrium

## An example

*Cost Matrix*

| | | Company 1 | |
|---|---|---|---|
| | | $M_1$ | $M_2$ |
| Company 2 | $M_1$ | 4 , 4 | 3 , 2 |
| | $M_2$ | 2 , 3 | 1 , 1 |

$$J_1(x_1, x_{-1}) = \begin{cases} 4 & \text{if } x_1 = M_1, x_{-1} = M_1 \\ 2 & \text{if } x_1 = M_1, x_{-1} = M_2 \\ 3 & \text{if } x_1 = M_2, x_{-1} = M_1 \\ 1 & \text{if } x_1 = M_2, x_{-1} = M_2 \end{cases}$$

$$J_2(x_2, x_{-2}) = \begin{cases} 4 & \text{if } x_2 = M_1, x_{-2} = M_1 \\ 2 & \text{if } x_2 = M_1, x_{-2} = M_2 \\ 3 & \text{if } x_2 = M_2, x_{-2} = M_1 \\ 1 & \text{if } x_2 = M_2, x_{-2} = M_2 \end{cases}$$

# Nash Equilibrium

## An example

*Cost Matrix*

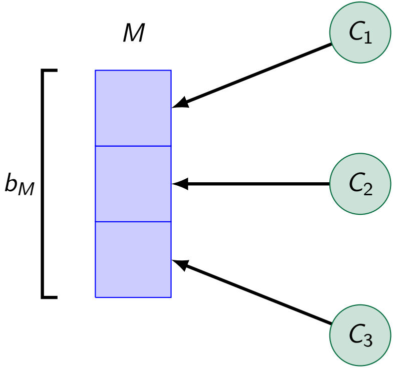
Company 1 (columns), Company 2 (rows)

| | $M_1$ | $M_2$ |
|---|---|---|
| $M_1$ | 4 , 4 | 3 , 2 |
| $M_2$ | 2 , 3 | 1 , 1 |

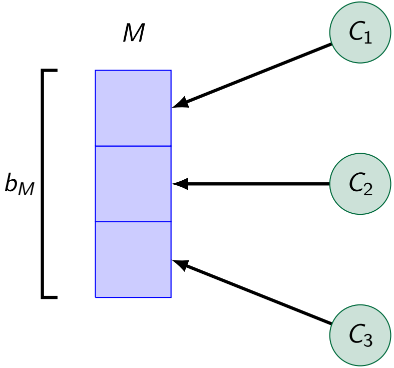
$$J_1(x_1, x_{-1}) = \begin{cases} 4 & \text{if } x_1 = M_1, x_{-1} = M_1 \\ 2 & \text{if } x_1 = M_1, x_{-1} = M_2 \\ 3 & \text{if } x_1 = M_2, x_{-1} = M_1 \\ 1 & \text{if } x_1 = M_2, x_{-1} = M_2 \end{cases}$$

$$J_2(x_2, x_{-2}) = \begin{cases} 4 & \text{if } x_2 = M_1, x_{-2} = M_1 \\ 2 & \text{if } x_2 = M_1, x_{-2} = M_2 \\ 3 & \text{if } x_2 = M_2, x_{-2} = M_1 \\ 1 & \text{if } x_2 = M_2, x_{-2} = M_2 \end{cases}$$

# Shared Constraints

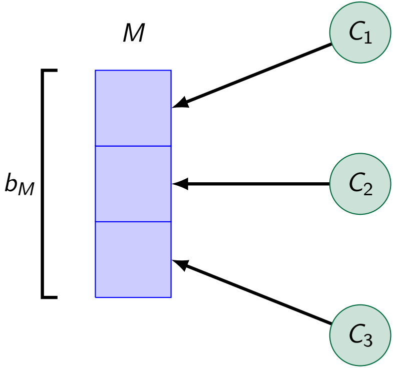


# Shared Constraints




$$x_1 + x_2 + x_3 \leq b_M$$

# Shared Constraints




$$x_1 + x_2 + x_3 \leq b_M$$

$$\Downarrow$$

$$A\boldsymbol{x} \leq b$$

# Generalized Nash Equilibrium

- Set of agents: $\mathcal{I} = \{1, \ldots, N\}$

  $\Rightarrow$ Companies

- Decision variable: $x_i \in \Omega_i$

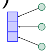
  $\Rightarrow$ Quantity of Energy

- Objective Function: $J_i(x_i, \boldsymbol{x}_{-i})$

  $\Rightarrow$ Cost - Profit

- Shared Constraints: $x_i \in \mathcal{X}(\boldsymbol{x}_{-i})$

  $\Rightarrow$ Market bounds



# Generalized Nash Equilibrium

- Set of agents: $\mathcal{I} = \{1, \ldots, N\}$

  $\Rightarrow$ Companies

- Decision variable: $x_i \in \Omega_i$

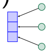
  $\Rightarrow$ Quantity of Energy

- Objective Function: $J_i(x_i, \boldsymbol{x}_{-i})$

  $\Rightarrow$ Cost - Profit

- Shared Constraints: $x_i \in \mathcal{X}(\boldsymbol{x}_{-i})$

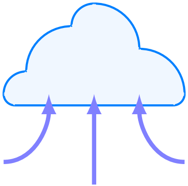
  $\Rightarrow$ Market bounds



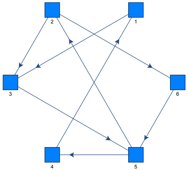
$$\forall i \in \mathcal{I} \quad \begin{cases} \min & J_i(x_i, \boldsymbol{x}_{-i}) \\ \text{s.t.} & A\boldsymbol{x} \leq b \end{cases}$$

# Cloud Storage Model

Classical



Decentralized



# CLOUD STORAGE MODEL

- Consider a set $\mathcal{I}$ of units where user $i$
  - need to store externally a back up of his data ($\alpha_i$)
  - can offer space available to other connected units ($\beta_i$)
- Units are connected through a network $\mathcal{G} = (\mathcal{I}, \mathcal{E})$ where $(i, j) \in \mathcal{E}$ iff unit $i$ is allowed to storage data in unit $j$

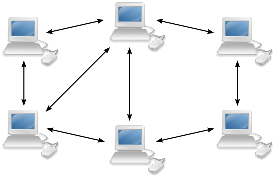


FIGURE: Network $\mathcal{G}$

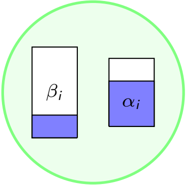



FIGURE: User $i$

# Allocation

Allocation state: any matrix $W \in \mathbb{N}^{\mathcal{I}\times\mathcal{I}}$, $W_{ij}$ quantity allocated in $j$ by $i$, such that

(Graph Constraint) $W_{ij} \geq 0$ for all $i, j$ and $W_{ij} = 0$ if $(i,j) \notin \mathcal{E}$.

(Allocation) $W^i := \sum_{j\in\mathcal{I}} W_{ij} \leq \alpha_i$ for all $i \in \mathcal{I}$.

(Storage Limitation) $W_j := \sum_{i\in\mathcal{I}} W_{ij} \leq \beta_j$ for all $j \in \mathcal{I}$.

# Allocation Game

- Set of agents: $\mathcal{I} = \{1, \ldots, N\}$
  $\Rightarrow$ Users
- Decision variable: $(W^i)$ $x_i \in \Omega_i$
  $\Rightarrow$ Data to be allocated $(\alpha_i)$
- Objective Function: $J_i(x_i, \boldsymbol{x}_{-i}) = J(W_i, W_{-i})$
  $\Rightarrow$ Congestion
- Shared Constraints: $A\boldsymbol{x} \leq \beta$
  $\Rightarrow$ Storage bounds

# Allocation Game

- Set of agents: $\mathcal{I} = \{1, \dots, N\}$
  $\Rightarrow$ Users
- Decision variable: $(W^i)$ $x_i \in \Omega_i$
  $\Rightarrow$ Data to be allocated $(\alpha_i)$
- Objective Function: $J_i(x_i, \boldsymbol{x}_{-i}) = J(W_i, W_{-i})$
  $\Rightarrow$ Congestion
- Shared Constraints: $A\boldsymbol{x} \leq \beta$
  $\Rightarrow$ Storage bounds

$$\forall i \in \mathcal{I} \quad \begin{cases} \min & J_i(x_i, \boldsymbol{x}_{-i}) \\ \text{s.t.} & A\boldsymbol{x} \leq \beta \end{cases}$$

# Connection with Markov Chains

## Allocation Algorithm

Poisson Process: with activation rates $\nu_i^{act}$

- Allocation Move: $W' = W + e_{ij}$
- Distribution Move: $W' = W + (e_{ij'} - e_{ij})$

# Connection with Markov Chains

## Allocation Algorithm

Poisson Process: with activation rates $\nu_i^{act}$

- Allocation Move: $W' = W + e_{ij}$
- Distribution Move: $W' = W + (e_{ij'} - e_{ij})$

# Connection with Markov Chains

## Allocation Algorithm

Poisson Process: with activation rates $\nu_i^{act}$

- Allocation Move: $W' = W + e_{ij}$
- Distribution Move: $W' = W + (e_{ij'} - e_{ij})$

# Connection with Markov Chains

## Allocation Algorithm

Poisson Process: with activation rates $\nu_i^{act}$

- Allocation Move: $W' = W + e_{ij}$ with probability $p^{all}$
- Distribution Move: $W' = W + (e_{ij'} - e_{ij})$ with probability $p^{dist}$

# Connection with Markov Chains

## Allocation Algorithm

Poisson Process: with activation rates $\nu_i^{act}$

- Allocation Move: $W' = W + e_{ij}$ with probability $p^{all}$
- Distribution Move: $W' = W + (e_{ij'} - e_{ij})$ with probability $p^{dist}$

$\Rightarrow$ Markov chain!
Guess what are the absorbing states?

Franci, B., Fagnani, F. (2018). A game theoretic approach to a network allocation problem. arXiv preprint arXiv:1809.07637

# Zero-Sum Games and Machine Learning

"Always include a work of art in your presentation"
*Francesco Bullo*

# Zero-Sum Games and Machine Learning

Tubingen, Germany

CycleGAN

# Zero-Sum Games and Machine Learning

Tubingen, Germany

VanGogh, The Starry Night

CycleGAN

[1] https://github.com/Adi-iitd/AI-Art

# Generative Adversarial Networks

Generator

Discriminator

# GENERATIVE ADVERSARIAL NETWORKS

## Generator



## Discriminator

# Generative Adversarial Networks

Generator

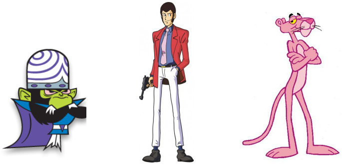


Discriminator

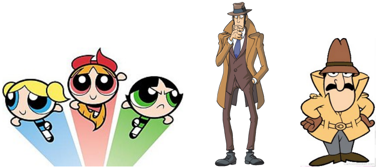


# Generative Adversarial Networks

## Generator



## Discriminator

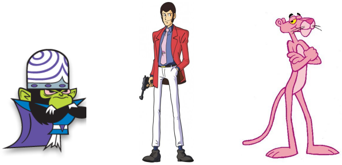


$\Rightarrow$ Neural Network

- Random noise: $z \sim p_z$
  Parameters: $x_g \in \Omega_g$
- Output $g(z, x_g) \in \mathbb{R}^q$ (image)

# Generative Adversarial Networks

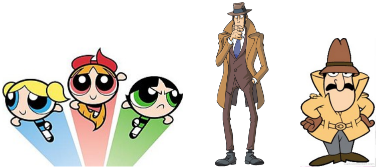
## Generator



$\Rightarrow$ Neural Network

- Random noise: $z \sim p_z$
  Parameters: $x_g \in \Omega_g$
- Output $g(z, x_g) \in \mathbb{R}^q$ (image)

## Discriminator

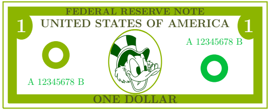


$\Rightarrow$ Neural Network

- Input $v$
  Parameters: $x_d \in \Omega_d$
- Output $d(v, x_d) \in [0, 1]$ (prob)

# GAN game


—




Zero-Sum game

Discriminator: $\mathbb{J}_{\mathrm{d}}(x_{\mathrm{g}}, x_{\mathrm{d}}) = \mathbb{E}[\phi(d(\cdot, x_{\mathrm{d}}))] - \mathbb{E}[\phi(d(g(\cdot, x_{\mathrm{g}}), x_{\mathrm{d}}))]$

Generator: $\mathbb{J}_{\mathrm{g}}(x_{\mathrm{g}}, x_{\mathrm{d}}) = -\mathbb{J}_{\mathrm{d}}(x_{\mathrm{g}}, x_{\mathrm{d}})$

Problem: $\min_{x_{\mathrm{g}}} \max_{x_{\mathrm{d}}} \mathbb{J}_{\mathrm{d}}(x_{\mathrm{g}}, x_{\mathrm{d}})$

# GANs Game

- Set of agents: $\mathcal{I} = \{1, \ldots, N\}$, $N = 2$
  $\Rightarrow$ Neural Networks
- Decision variable: $x_i \in \Omega_i$
  $\Rightarrow$ Parameters
- Objective Function: $\mathbb{J}_i(x_i, \boldsymbol{x}_{-i})$
  $\Rightarrow$ Distance from real currency

# GANs Game

- Set of agents: $\mathcal{I} = \{1, \ldots, N\}$, $N = 2$
  $\Rightarrow$ Neural Networks
- Decision variable: $x_i \in \Omega_i$
  $\Rightarrow$ Parameters
- Objective Function: $\mathbb{J}_i(x_i, \boldsymbol{x}_{-i})$
  $\Rightarrow$ Distance from real currency

$$\begin{cases} \min_{x_\mathrm{g}} \max_{x_\mathrm{d}} \mathbb{J}_\mathrm{d}(x_\mathrm{g}, x_\mathrm{d}) \\ \text{s.t.} \quad x_i \in \Omega_i \end{cases}$$

Franci, B., Grammatico, S. (2020). Training Generative Adversarial Networks via stochastic Nash games. arXiv preprint arXiv:2010.10013.

# GANs Game

- Set of agents: $\mathcal{I} = \{1, \ldots, N\}$, $N = 2$
  $\Rightarrow$ Neural Networks
- Decision variable: $x_i \in \Omega_i$
  $\Rightarrow$ Parameters
- Objective Function: $\mathbb{J}_i(x_i, \boldsymbol{x}_{-i})$
  $\Rightarrow$ Distance from real currency

$$\begin{cases} \min_{x_\mathrm{g}} \max_{x_\mathrm{d}} \mathbb{J}_\mathrm{d}(x_\mathrm{g}, x_\mathrm{d}) \\ \text{s.t.} \quad x_i \in \Omega_i \end{cases}$$

$$\forall i \in \mathcal{I} \quad \begin{cases} \min_{x_i} \mathbb{J}_i(x_i, x_{-i}) \\ \text{s.t.} \quad x_i \in \Omega_i \end{cases}$$

Franci, B., Grammatico, S. (2020). Training Generative Adversarial Networks via stochastic Nash games. arXiv preprint arXiv:2010.10013.

# Exercise

Can you think of a game for:

- Ride-Hail market
- Smart-Building Optimization
- Penalty Kicks in Football

# Exercise

Can you think of a game for:

- Ride-Hail market
- Smart-Building Optimization
- Penalty Kicks in Football

Fabiani, F., Franci, B. (2022). A stochastic generalized Nash equilibrium model for platforms competition in the ride-hail market. arXiv preprint. arXiv:2203.15412.

- Chiappori, P. A., Levitt, S., Groseclose, T. (2002). Testing mixed-strategy equilibria when players are heterogeneous: The case of penalty kicks in soccer. American Economic Review, 92(4), 1138-1151. doi: 10.1257/00028280260344678.